Software Testing

Decision Tables

Decision table

- A **decision table** is a good way to deal with different combination inputs with their associated outputs and also called cause-effect table.

Decision table

- **Decision table testing is black box test design technique to determine the test scenarios**

Decision table

Step A (RE phase)
“We use decision tables to make sure we understand all business rules correctly.”
Step B (Testing phase)
“We use the same decision table to ensure we test every rule systematically.”
Key message to students:
“Requirements define the rules; testing ensures every rule is exercised.”

Big Picture Relationship

Technique	What it models	Main idea
Equivalence Partitioning (EP)	Input domains	Divide inputs into valid/invalid groups
Boundary Value Analysis (BVA)	Input boundaries	Test edges of partitions
Decision Tables (DT)	Condition combinations	Test logical rules across multiple conditions

EP and BVA focus on **single input variables**, while decision tables focus on **multiple interacting conditions**.

Decision table

- **Example 1:** Decision Base Table for Login Screen
- Let's create a decision table for a login screen.

Example

Example

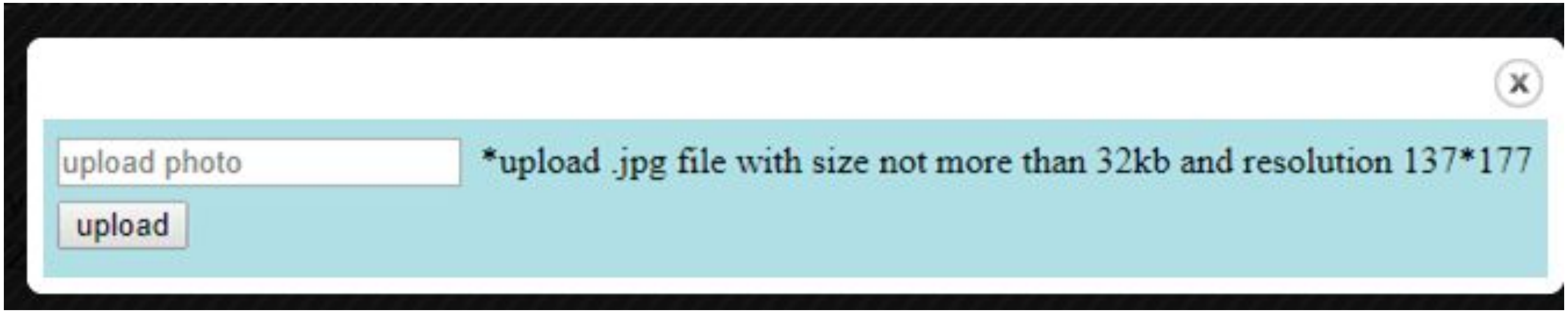
Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Username (T/F)	F	T	F	T
Password (T/F)	F	F	T	T
Output (E/H)	E	E	E	H

Legend:
T – Correct username/password
F – Wrong username/password
E – Error message is displayed
H – Home screen is displayed

Example 2:

- Decision Base Table for Upload Screen, Now consider a dialogue box which will ask the user to upload photo with certain conditions like –
- You can upload only '.jpg' format image
- file size less than 32kb
- resolution 137*177.

Example 2:


upload photo
*upload .jpg file with size not more than 32kb and resolution 137*177
upload

Decision table

Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
.jpg	.jpg	.jpg	.jpg	Not .jpg	Not .jpg	Not .jpg	Not .jpg
Less than 32kb	Less than 32kb	More than 32kb	More than 32kb	Less than 32kb	Less than 32kb	More than 32kb	More than 32kb
137*177	Not 137*177	137*177	Not 137*177	137*177	Not 137*177	137*177	Not 137*177
Photo uploaded	Error message resolution mismatch	Error message size mismatch	Error message size and resolution mismatch	Error message for format mismatch	Error message format and resolution mismatch	Error message for format and size mismatch	Error message for format, size, and resolution mismatch

Example #3

- Let us consider the following description of a payment procedure. Consultants working for more than 40 hours per week are paid at their hourly rate for the first 40 hours and at two times their hourly rate for subsequent hours. Consultants working for less than 40 hours per week are paid for the hours worked. at their hourly rates and an absence report is produced. Permanent workers working for less than 40 hours a week are paid their salary and an absence report is produced. Permanent workers working for more and equal to 40 hours a week are paid their salary.
- We need to describe the above payment procedure using a decision table and generate test cases from the table.

Cause and Effects

- *C1: Permanent workers*
- *C2: Worked < 40 hours*
- *C3: Worked exactly 40 hours*
- *C4: Worked > 40 hours*
- *E1: Pay salary*
- *E2: Produce an absence report*
- *E3: Pay hourly rate*
- *E4: Pay 2 × hourly rate*

Decision Table

		Rules or Combinations															
Conditions	Values	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
C_1	Y, N	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N	N
C_2	Y, N	Y	Y	Y	Y	N	N	N	N	Y	Y	Y	Y	N	N	N	N
C_3	Y, N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N
C_4	Y, N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
Effects																	
E_1																	
E_2																	
E_4																	
E_4																	

Reduced Table

		Rules or Combinations					
Conditions	Values	1	2	3	4	5	6
C_1	Y, N	Y	Y	N	N	N	N
C_2	Y, N	Y	N	Y	N	N	N
C_3	Y, N	—	—	—	Y	N	N
C_4	Y, N	—	—	—	—	Y	N
Effects							
E_1							
E_2							
E_4							
E_4							

Table With Check Sum

Conditions	Values	Rules or Combinations 1	2	3	4	5	6
C_1	Y, N	Y	Y	N	N	N	N
C_2	Y, N	Y	N	Y	N	N	N
C_3	Y, N	—	—	—	Y	N	N
C_4	Y, N	—	—	—	—	Y	N
Effects							
E_1							
E_1		1	1				
E_2		2		2			
E_4				1	1	1	
E_4						2	
Checksum	16	4	4	4	2	1	1